ETEC510 DESIGN PROJECT

PHASE 2: CURRICULUM USER GUIDE (DRAFT)

Critical Thinking Tool-kit for High-Schoolers

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I. Introduction & Goals

We present our Curriculum User Guide, which provides a high-level overview of developing a Gamified Critical Thinking Skills Course for educators, aligned with the BC curriculum and adaptable to other provincial curricula.

Our stated goals are to engage learners in a constructivist, relevant, differentiated, and design-based learning environment. The plan is to embed in the curricular competencies of the BC Curriculum and the various aspects of the Core Competencies, fostering a community of learning and the development of a curriculum for 21st-century learners.

More specifically, we plan to provide a critical thinking toolkit for grade 10 students to understand and use. Our aim is for students to use these skills in real-life situations by providing a gamified knowledge progression course. At the end, students are expected to use these skills to develop cases of their own and challenge each other to develop true proficiency in these skills.

By creating this short course, teachers will be able to integrate the material into their curriculum planning and engage students in meaningful and purposeful ways. The tool can be integrated into either e-learning or in-person career studies courses, at the teacher's discretion.

II. Pedagogy – Participatory Culture

Our toolkit is grounded in a participatory cultural pedagogy that supports students in developing critical thinking through situated practice. Situated practice is presented to students within a series of domains relevant to tenth grade students such as relationships, academic decision-making, and career life planning. Within these contexts, students are given the opportunity to analyze evidence, form judgments, and most importantly use their critical thinking skills to mirror real-world participation.

Drawing on our collective understanding of Henry Jenkins et al.'s (2009) work on participatory culture, the toolkit is designed to give students opportunities to meaningfully exercise their agency within an engaging learning environment. **Our goal is to provide "a safe space within which they can master the skills they need as citizens and consumers as they learn to parse messages from self-interested parties, and where they can separate fact from falsehood as they begin to experiment with new forms of creative expression and community participation"** (Jenkins, 2009, p. 24).

<u>Our toolkit supports participatory culture by:</u>

1. **Allowing students to exercise their agency**: Learners are given a space to intervene in scenarios, make decisions, and experience the outcomes of their choices.

2. **Valuing contribution:** Skill development takes time, and our design of this toolkit features badges that reward the progress the student makes. This offers a space for students to make mistakes in a low-stakes environment.

For educators, implementing this toolkit requires intentionally creating classroom conditions where participatory culture can emerge. This aligns with Jenkins et al.'s (2009) call for educators to provide spaces where young people can access the skills and experiences needed to become full participants.

In practice, supporting participatory culture means reducing barriers to engagement and expression. In accordance with Jenkins et al.'s (2009) understanding of participatory culture, this toolkit positions educators as guides who support students as they navigate scenarios grounded in their own lived contexts.

III. Curriculum Connection – BC curriculum connection

This section maps the toolkit to the BC Core Competencies framework, demonstrating how it supports the transferable capacities BC expects students to develop across all courses and grade levels. While Career Life Education 10 is our proposed primary delivery context, the Core Competencies framing makes the case for the toolkit's broader curricular value across subject areas, grade levels, and provinces. Cross-provincial equivalencies are available through the CERIC Career Curriculum guide (careerwise.ceric.ca).

The BC Core Competencies: An Overview

BC's Core Competencies are transferable capacities that students build across every subject and grade level, from kindergarten through graduation. They are not a separate course — they are woven into all learning, documented through teacher observation and student self-reflection, and reported on as part of the standard assessment cycle.

BC identifies three Core Competency areas:

- **Communication** — connecting with others, acquiring and presenting information, collaborating
- **Thinking** — subdivided into Creative Thinking and Critical & Reflective Thinking
- **Personal & Social** — subdivided into Personal Awareness & Responsibility and Social Awareness & Responsibility

Because Core Competencies cut across all subjects, any tool that develops them has curricular legitimacy in multiple classrooms — not just one course. A critical thinking toolkit that explicitly targets these competencies can be used in English Language Arts, Social Studies, Career Education, or any other subject where teachers are looking for structured ways to document and develop student thinking.

Key point: *Core Competencies are assessed through observable student behavior, not tests. This means a gamified, activity-based toolkit is a particularly well-suited vehicle for developing and documenting them*

Mapping the Three Lenses to BC Core Competencies

Shafer's (2025) three-lens critical thinking framework — pragmatic, dispositional, and transformational — maps directly onto BC's Core Competency clusters. Each badge students earn in the toolkit corresponds to a lens and to a distinct competency area. The third column outlines possible strategies for how the toolkit could develop each competency.

Badge / Lens	BC Core Competency	Possible Strategies
Pragmatic (Reasoning)	Critical & Reflective Thinking — questioning assumptions, analyzing evidence, forming and defending judgments	• Show students two versions of the same information and ask them to decide which to trust • Help students see how outside forces — ads, algorithms, trends — shape what information they see • Ask students to follow a claim back to where it started
Dispositional (Reflection)	Personal Awareness & Responsibility — self-regulating, reflecting on learning, developing identity and values	• Ask students to explain how they reached their decision, not just what they decided • Show students how an expert approached the same problem, without framing it as correction • Let students track their own progress so they can notice how their thinking is changing
Transformational (Agency)	Social Awareness & Responsibility — contributing to community, considering consequences for others, valuing diverse perspectives	• Put students in situations where their choices affect others, not just themselves • Have students work together to reach a decision they each have to defend • Ask students to create something — a challenge, an argument, a case — that a peer will engage with
All lenses	Communication — explaining and justifying thinking, collaborating, acquiring and presenting information	• Let students speak their thinking out loud before they have to write it down • Have peers identify one specific strength in each other's reasoning, not just whether they agree • Use multiplayer tasks that require students to negotiate and communicate across different perspectives

Possible Area to Explore: Aligning with BC Proficiency Scale

BC describes Core Competency development using four proficiency levels: Emerging, Developing, Proficient, and Extending. One direction worth exploring is whether the toolkit badge and progression system could be mapped to this scale with students moving from structured, scaffolded tasks toward independent application, roughly mirroring the shift from Emerging to Extending. The table below sketches out what this might look like and is offered as a starting point for further discussion rather than a finalized design decision.

Proficiency Level	What It Looks Like in BC's Framework	Where the Toolkit Addresses This
Emerging	Student is beginning to identify and apply the competency with significant support	Early guided missions with visible frameworks, hints, and expert modeling
Developing	Student applies the competency with some support and is growing in consistency	Mid-level missions with scaffolding that fades as students demonstrate confidence
Proficient	Student applies the competency independently and consistently in familiar contexts	Badge completion tasks requiring independent judgment across a full case
Extending	Student applies the competency in new contexts and supports the learning of others	Sandbox phase and peer creation tasks where students design challenges for classmates

Cross-Curricular Delivery Options

Because Core Competencies belong to every subject, this toolkit has a legitimate home in multiple classrooms. The table below identifies subject areas where teachers could integrate it, and the primary competency each context would emphasize.

Subject Area	Primary Competency Emphasis	Natural Entry Point
Career Life Education 10	Personal & Social; Communication	Self-Assessment or Workplace Readiness units; digital footprint content
English Language Arts 10	Critical & Reflective Thinking; Communication	Media literacy, persuasion, and source analysis units

Social Studies 10	Critical & Reflective Thinking; Social Awareness & Responsibility	Civic participation, media and democracy, or current events units
Applied Design, Skills & Technologies	Creative Thinking; Communication	Design thinking, iterative problem-solving, or digital literacy modules
Any subject (advisory/homeroom)	All Core Competencies	Stand-alone module delivered during advisory, flex, or study blocks

A Note on Pedagogy: Cognitive Apprenticeship as One Possible Framework

Teachers implementing this toolkit may find cognitive apprenticeship (Collins, Brown & Newman, 1989) a useful lens. The framework proposes making expert thinking visible to learners through modeling, coaching, scaffolding, and gradually releasing responsibility — a sequence that maps naturally onto the toolkit's own progression from guided missions to open sandbox. This is one possible implementation approach, not a requirement; teachers should adapt the tool to their own pedagogical context and subject area.

Reference: *Collins, A., Brown, J. S., & Newman, S. E. (1989). Cognitive apprenticeship: Teaching the crafts of reading, writing, and mathematics. In L. B. Resnick (Ed.), Knowing, learning, and instruction (pp. 453–494). Lawrence Erlbaum.*

Cross-Provincial Applicability

The Core Competencies framework is BC-specific, but the competencies themselves - critical thinking, communication, personal and social responsibility - are shared goals across Canadian provincial curricula. The three-badge structure (reasoning, reflection, agency) maps broadly onto career development, personal planning, and media literacy courses nationwide. Teachers outside BC can use the CERIC guide (careerwise.ceric.ca) to identify the closest local equivalents.

IV. Technology Requirements

Moodle is a Learning Management System (LMS) that will be used to house the course in its entirety. Educators will be able to develop the course through this LMS, and tailor the content for their students.

https://moodle.org/

Moodle is able to handle:

- Course Setup
- Enrolment
- Grading
- Progress
- Access & Security

https://docs.moodle.org/501/en/Teacher_quick_guide

Genially is the gamification engine behind the courses' learning objects, which appropriately syncs with the LMS to provide the intended engagement with students.

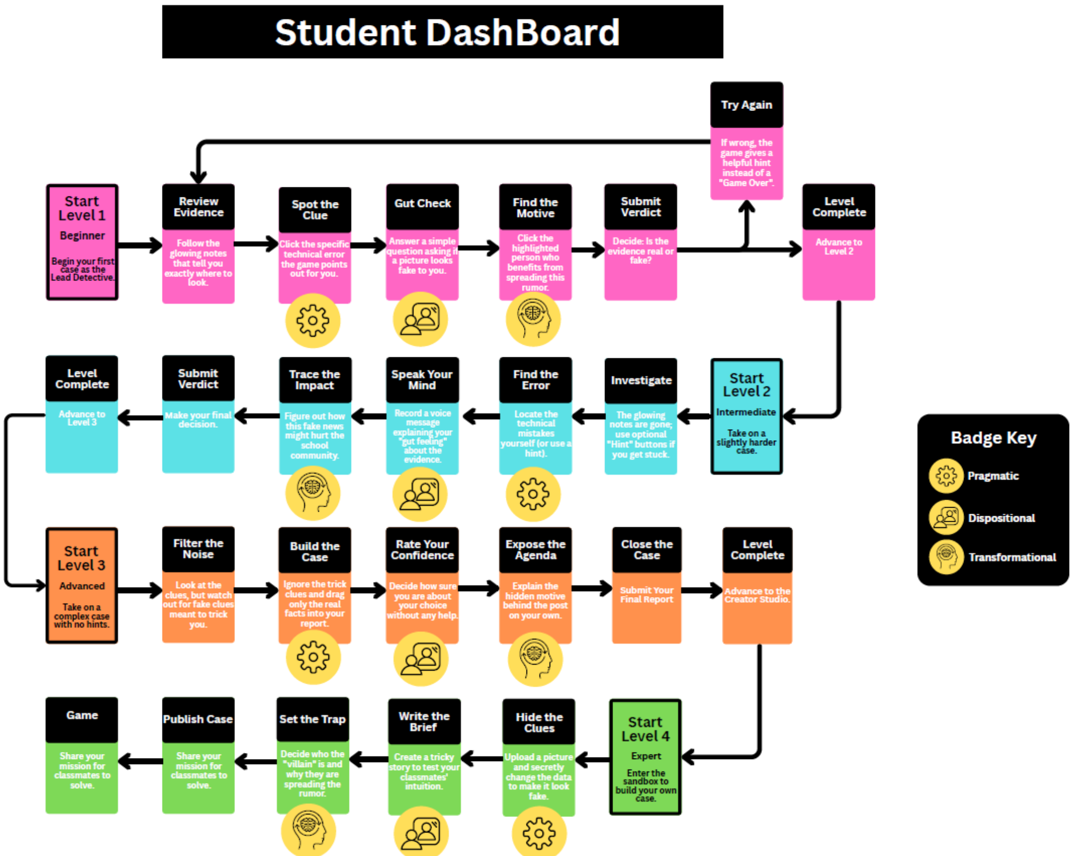
https://genially.com/

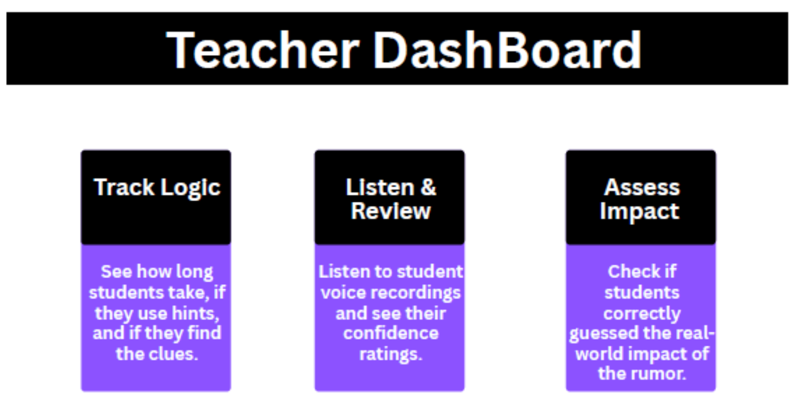
Genially will be used for:

- Interactive Content
- Animations and Scenario Building
- Infographics & Presentation

https://genially.com/features/

V. Visual Progression User Guide





Visual Progression User Guide: The Learner's Journey

The Digital Detective toolkit is designed as a dynamic, interactive learning environment where progression is based strictly on cognitive complexity. In order to advance, each mission requires the student to successfully fulfill and earn three specific critical thinking badges Pragmatic, Dispositional, and Transformational at their current level of complexity.

Phase 1: Beginner

In the initial stage, students log into the Moodle Learning Management System and launch the interactive Genially module. At this beginner level, the interface provides significant expert modeling, explicitly guiding students toward earning their first set of badges.

Mechanics & Badge Acquisition: Students are introduced to the Evidence Board, where glowing "Detective's Notes" explicitly direct their attention.

- **Earning the Pragmatic Badge:** Students must click explicitly highlighted technical clues, such as a flagged software tag in the metadata log.
- **Earning the Dispositional Badge:** Students are prompted with a highly scaffolded "gut check" question, asking them to select whether a visual anomaly (like six fingers on a hand) feels unnatural.
- **Earning the Transformational Badge:** The UI will specifically prompt a student to use the Transformational lens to ask who benefits from a piece of media being shared. Glowing notes guide students to identify the specific beneficiary of a fake post, such as a brand-new social media account seeking engagement.

Participatory Culture Integration (Safe Failure): To foster a safe space for skill mastery without the fear of penalization, this level features a built-in "Redemption Mechanic." If a student makes an incorrect judgment or submits the wrong final verdict, they are not met with a "Game Over." Instead, they are gently redirected back to the evidence to try again, reinforcing that mistakes are a helpful and necessary part of the learning process.

Phase 2: Intermediate

As students enter the intermediate phase, the direct instructional scaffolding begins to fade, requiring more active cognitive engagement.

Mechanics & Badge Acquisition: The explicit "Detective's Notes" are removed. In their place, the UI offers optional "Hint" buttons. Students must now begin to rely on their own internal reasoning to apply Shafer's three thematic lenses.

- **Earning the Pragmatic Badge:** Students must locate visual and technical anomalies using the optional, user-triggered "Hint" buttons if they get stuck.
- **Earning the Dispositional Badge:** To support diverse communication styles and reduce barriers to expression, students must use a "Record Voice Memo" feature before submitting their final verdict. They earn this badge by speaking their reasoning aloud, allowing them to articulate their thoughts authentically without being hindered by spelling-related self-consciousness.
- **Earning the Transformational Badge:** Students must analyze a scenario and identify the broader social consequences of the media spreading, moving beyond just identifying the original poster.

Phase 3: Advanced

In the advanced phase, the training wheels are entirely removed. Students are expected to apply the BC Core Competencies independently in the presence of distracting information.

Mechanics & Badge Acquisition: The Evidence Board introduces "Red Herrings" pieces of evidence that are irrelevant or purposefully misleading.

- **Earning the Pragmatic Badge:** Students must independently filter out irrelevant clues, successfully drag the relevant evidence into their "Case Closed" report, and justify their reliance on the correct technical facts.
- **Earning the Dispositional Badge:** Students rate their own confidence level on a sliding scale and defend their stance without any prompting or hints.
- **Earning the Transformational Badge:** Students independently predict the community impact and hidden agendas behind the media campaign.

Critical Application: Earning badges at this level signifies a strong, independent grasp of digital media literacy, as students have successfully and independently articulated their Pragmatic , Dispositional, and Transformational justifications.

Phase 4: Expert

The final phase of the flowchart represents the pinnacle of Jenkins' (2009) Participatory Culture. The student transitions from a consumer of the game to a creator.

Mechanics & Badge Application: The UI shifts to a "Case Creator" level editor. Students use a sandbox approach to upload their own media, plant visual anomalies, manipulate metadata, and assign badge triggers.

- **Applying the Pragmatic Skill:** Students intentionally plant logical anomalies or alter metadata in the media they upload.
- **Applying the Dispositional Skill:** Students write the case briefing to intentionally test the intuition and biases of their peers.
- **Applying the Transformational Skill:** Students design the overarching social scenario and hidden motives that their classmates will need to uncover.

Social Learning: By publishing these custom cases for their peers to solve, students not only demonstrate extended mastery of the subject matter but also actively support the learning of their classmates, fulfilling the highest tier of the BC Proficiency Scale. This grants them the ultimate "Lead Detective" status.

The Backend: Teacher Dashboard

Running parallel to the learner's journey is the Teacher Assessment Dashboard. As the student interacts with the Genially modules, data is securely pushed to the educator. The dashboard visualizes the student's badge acquisition, tracks time spent on evidence, logs hint usage, and stores their recorded voice memos. This provides the educator with observable, concrete data points to apply grading rubrics accurately, ensuring the gamified experience remains deeply tied to measurable curricular outcomes.

V. Assessment

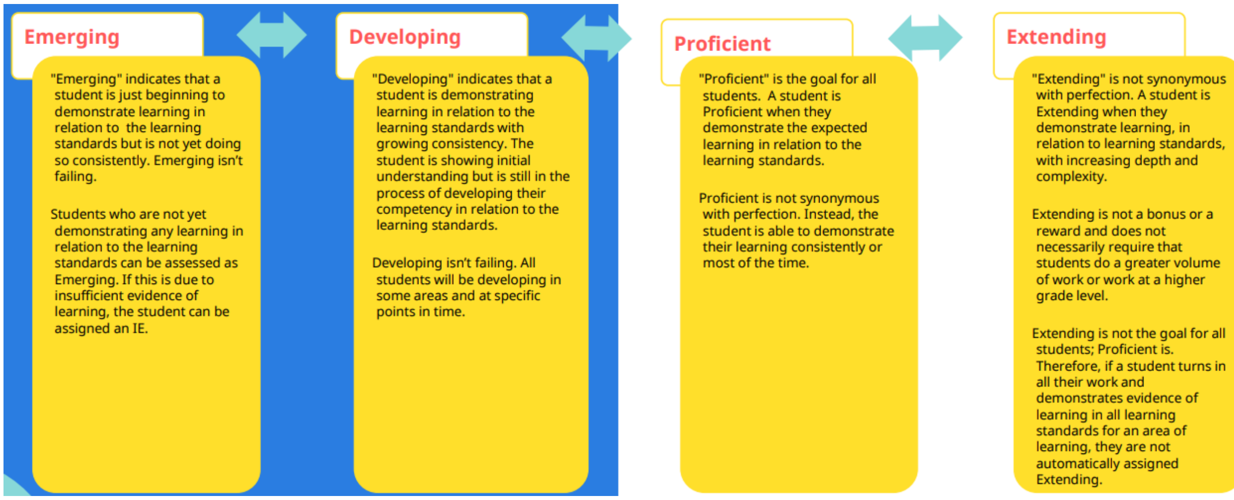
Formative and Summative Tasks

Students will be assessed through formative and summative tasks. The priority of formative assessments will be to check in on learning to guide students on their journey through the material. The tasks can include informal observations and low-stakes quizzes. Summative assessments are meant to assess final learning to examine what students have taken from the module and what skills they can demonstrate. Below is a list of suggested assessments.

Formative Assessment	Summative Assessment
• Activities between earning badges for practice • Progress tracking to see whether or not students are on the right path • Observations of student conversations and collaborations to come to new conclusions and perspectives • Informal check-ins – short quizzes for understanding	• Ability to navigate through summative tasks to gain badges through explaining reasoning • Predicting hidden agendas • Voice Memo Justifications • Analysis of scenarios • Participation in Case Studies and activities

Grading

Assessment is based on the BC Proficiency Scale rather than quantitative scoring. Instead of accumulating points, educators will use rubrics to evaluate qualitative observations of a student's skill acquisition. For instance, a student’s grade is determined by their ability to demonstrate reasoned thinking in the gamified summative tasks. When using this tool, educators are encouraged to grade based on the learning tasks in levels 3 and 4. This ensures that students have been given the opportunity to practice in earlier levels to support authentic assessment.



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Sample Rubrics

Students will be graded based on the guidance of a rubric to assess their proficiency. Educators using this tool can create their own rubrics, but below are some samples of the expected outcomes of curricular competencies across a variety of subject areas.

Career Education 10

	Emerging	Developing	Proficient	Extending
Identify risks and appreciate benefits associated with personal and public digital footprints	Identifies basic consequences of digital footprint with significant support using the transformational lens.	Identifies consequences of digital footprint with some support using the transformational lens.	Identifies consequences of their digital footprints on themselves and others using the transformational lens	Identifies consequences of their digital footprints in complex scenarios using the transformational lens.
Identify career-life challenges and opportunities, and generate and apply strategies	Identifies basic motives of digital media with significant support using the pragmatic lens.	Identifies underlying motives of digital media with some support using the pragmatic lens.	Identifies underlying motives of digital media using the pragmatic lens.	Identifies a variety of underlying motives of digital media using the pragmatic lens.

New Media (English Language Arts) 10

	Emerging	Developing	Proficient	Extending

Think critically, creatively, and reflectively to explore ideas within, between, and beyond texts	Reflects on their own thinking process with significant support using the dispositional lens.	Reflects on their own thinking processes with some support using the dispositional lens.	Reflects on their own thinking processes in familiar contexts using the dispositional lens.	Reflects on their own thinking processes in complex contexts using the dispositional lens.
Respectfully exchange ideas and viewpoints from diverse perspectives to build shared understanding and extend thinking	Explains their positioning with minimal detail begins to listen to opposing perspectives using the transformational lens.	Explains their positioning on familiar contexts with support and listens to opposing perspectives using the transformational lens.	Explains their positioning in familiar contexts with clarity and understands opposing perspectives using the transformational lens.	Explains their positioning in complex contexts with clarity and understands and can connect to opposing perspectives using the transformational lens.

Social Studies 10

	Emerging	Developing	Proficient	Extending
Make reasoned ethical judgments about actions in the past and present, and assess appropriate ways to remember and respond	Begins to apply reasoning on how outside forces can influence the presentation of information using the pragmatic lens.	Applies reasoning on how outside forces can influence the presentation of information with support using the pragmatic lens.	Applies reasoning on how outside forces can influence the presentation of information using the pragmatic lens.	Applies sophisticated reasoning on how outside forces can influence the presentation of complex information using the pragmatic lens.

VI. Summary

Youth are more engaged with digital media environments and must also develop important skills to participate critically and ethically. True to the participatory learning philosophy, this guide reinforces Jenkins' view: “a focus on expanding access to new technologies carries us only so far if we do not also foster the skills and cultural knowledge necessary to deploy those tools toward our own ends.” (Jenkins, 2009, p. 8)

This idea directly informs the goals of this course regarding ways to develop and reinforce students' critical thinking skills to carefully engage with digital and media consumption and participation.

The course will provide students with opportunities for lifelong learning across various domains, including relationships, career pathways, and educational planning. Through its development, the course will aim to incorporate gamification elements that foster a reflective and collaborative learning environment while effectively integrating digital technologies.

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